AAE 534 Spacecraft Electric Propulsion - Fall 2025
3 credits

1 Course Description

Spacecraft electric propulsion systems are intended to provide thrust for propelling spacecrafts in interplanetary missions, orbital maneuvers and attitude control. The course will start with reviewing material on mechanics and thermodynamics of propulsion, and identifying the niche occupied by the electric propulsion systems. The course will cover elements of plasma physics and electromagnetic theory essential for studying the electric propulsion systems. The core of the course will focus on studying various electric propulsion concepts which utilize electric power to generate thrust. Mechanisms of the utilization of the electric power to accelerate gas or plasma and produce thrust will be considered, including electrothermal, electrostatic, electromagnetic, and gasdynamic acceleration mechanisms.

2 Learning Resources and Reference Text

Brightspace Learning Management System: All course materials including course announcements, blank and filled lecture notes, homework assignments, etc. will be posted on the course Brightspace site.

Textbook: R.G. Jahn, *Physics of Electric Propulsion* New York: McGraw-Hill, 1968.

3 Course Learning Objectives

By the end of the semester, successful students will be able to:

- Demonstrate fundamental understanding of concepts and acceleration mechanisms utilized in electric propulsion systems (electrothermal, electrostatic, electromagnetic).
- Perform analysis of electrothermal accelerator at given frozen flow fraction
- Perform analysis of ion acceleration in space-charge limited electrostatic accelerators
- Perform basic analysis of plasma acceleration in crossed-field electromagnetic thruster
- Identify optimal electric propulsion system based on specific mission requirements

4 Assignments

4.1 Homework:

Homework will be assigned once a week (with a few exceptions) and will be due one week later. Homework assignments will be turned in electronically in Brightspace.

4.2 Mid-term exam

The mid-term exam will be held in October.

4.3 Final exam

The final exam will be held during Finals Week.

5 Evaluation

Course grading:	Homework	= 35%
	Mid-term exam	= 30%
	Final exam	= 35%
	TOTAL	= 100%

Grading schema: A+ (≥97), A (≥93), A- (≥90), B+(≥87), B (≥83), B- (≥80), C+ (≥77), C (≥73), C- (≥65), D (≥60), F (<60)